# Ajith Surya Thati

**Senior Platform & DevOps Engineer**

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## Summary

DevOps and Platform Engineer with 8+ years of experience supporting, automating, and optimizing production deployments on AWS, Kubernetes, and hybrid cloud platforms. Proven track record designing and maintaining CI/CD pipelines across Jenkins, Concourse, and GitHub Actions; automating infrastructure with Terraform, AWS CDK, and Ansible; and managing containerized microservices on Kubernetes/EKS. Experienced handling out-of-hours on-call support rota (PagerDuty), driving tooling migrations (Chef to Cinc, Cloudflare WAF, AL2023 Launch Templates), and integrating secure secrets management (AWS IRSA, External Secrets, Secrets Manager). Active developer of cross-platform desktop media utilities (FFmpeg) and full-stack cloud applications (Next.js, React, Firebase).

## Work Experience

**Flutter UK & Ireland (Sky Betting & Gaming)** Leeds, UK (Hybrid)
*Platform / DevOps Engineer* *June 2023 – Present*

- Manage, deploy, and maintain multi-cluster AWS EKS infrastructure and CI/CD pipelines (Jenkins, GoCD, Concourse, GitHub Actions) supporting high-traffic Free to Play betting and gaming products.
- Systematically executed multi-version in-place upgrades of production EKS clusters from **v1.26 through v1.31**; audited and verified add-ons (`VPC CNI`, `CoreDNS`, `kube-proxy`, `EBS CSI`, `Cluster Autoscaler`) to prevent control-plane incompatibility with zero customer downtime.
- Migrated production worker node fleets from legacy EC2 Launch Configurations to **Launch Templates on Amazon Linux 2023 (AL2023)** with zero service disruption.
- Deployed and maintained observability tools across AWS EKS using Helm, Terraform, Prometheus, Grafana, and Alertmanager; participate in primary out-of-hours on-call support rota (PagerDuty, JIRA) diagnosing cluster and service anomalies.
- Executed platform tooling migrations including Chef to Cinc, AWS WAF to Cloudflare WAF, and Concourse CI pipelines to GitHub Actions.
- Configured Amazon RDS instances, resolved complex network dependencies, and established connectivity between Network Load Balancers (NLB) and RDS using VPC Endpoint Services.
- Authored custom AWS Lambda functions to monitor RDS endpoints and dynamically update NLB target groups with active IP addresses; built Lambda functions to query CloudFront IP ranges and update S3 bucket policies.
- Designed reusable AWS CDK Custom Resources and Lambda automations to provision service-specific MySQL users with least-privilege permissions and store credentials in AWS Secrets Manager.
- Implemented secure pod access using IAM Roles for Service Accounts (IRSA) and External Secrets Operator, replacing legacy Vault and Kube2IAM integrations.
- Created Docker base images from scratch, customized base images from existing environments, and managed centralized container image repositories for product engineering squads.
- Evaluated and configured an internal AI agent toolkit across development teams; audited tool codebase for telemetry and supply-chain dependencies to verify zero data exfiltration prior to rollout.

**IBM** Remote / Hyderabad, India
*DevOps Engineer* *April 2021 – May 2023*

- Developed microservice onboarding automation tools using Python and Jenkins, streamlining creation and maintenance of build jobs and Kubernetes deployments.
- Managed containerized deployments using Docker and Kubernetes, authoring custom Dockerfiles, Docker Compose configurations, and Helm charts for CI/CD automation.
- Automated multi-environment infrastructure provisioning and configuration management using Terraform and Ansible Tower playbooks.
- Integrated automated code scanning and security vulnerability analysis into pipelines using Sonatype and Veracode, ensuring code quality and baseline compliance.
- Monitored and analysed system logs using Splunk dashboards to improve performance and enable proactive incident troubleshooting.
- Automated build and release tasks using Bash/Shell scripting and JSON templates across VSTS and Git pipelines; collaborated with teams on Git branching strategies and merge conflict resolution.
- Assisted in configuring Kafka environments for internal projects, including setup of brokers, topics, and schema registry using Kafka Connect.
- Supported Agile/Scrum delivery by managing JIRA sprint tracking, change management, and providing technical guidance to team members on DevOps best practices.

**Tech Mahindra** Hyderabad, India
*DevOps Engineer / Software Engineer* *Nov. 2016 – March 2021*

- Designed and deployed automated CI/CD pipelines using Jenkins, Git, Maven, Selenium, and JUnit, reducing software release cycle times by 50%.
- Deployed and managed AWS cloud infrastructure across EC2, VPC, S3, IAM, Route 53, and Lambda; automated server provisioning and Auto Scaling groups using Terraform and CloudFormation.
- Developed reusable Jenkins shared libraries and integrated pipeline status alerts with Slack and Microsoft Teams for real-time build notifications.
- Built and orchestrated containerized microservices using Docker, Kubernetes, Docker Swarm, and Docker Compose; automated container deployments with Ansible playbooks.
- Configured Classic and Application Load Balancers (ALB) with SSL/TLS certificates across staging and production environments for high-availability traffic distribution.
- Administered artifact repositories using JFrog Artifactory and Sonatype Nexus to manage application build dependencies and release packages.
- Implemented Istio service mesh for microservices traffic management, canary rollouts, and secure service-to-service communication.
- Developed PowerShell and Linux shell scripts for server housekeeping, automated backups, and operational health checks.
- Collaborated with cross-functional development, QA, and operations teams to coordinate smooth production releases and troubleshoot deployment issues.

## Technical Skills

**Cloud Infrastructure**: AWS (EKS, EC2, VPC, RDS, Lambda, NLB/ALB, CloudFront, S3, IAM/IRSA, Route53, Secrets Manager, SSM Parameter Store), GCP, Azure
**Containers & Orchestration**: Kubernetes, Docker, Helm, ArgoCD, Argo Rollouts, ISTIO Service Mesh, ECR, Docker Compose
**Infrastructure as Code (IaC)**: Terraform, AWS CDK (TypeScript), Ansible, Ansible Tower, AWS CloudFormation
**CI/CD & Automation**: Jenkins (Shared Libraries), GitHub Actions, Concourse CI, GoCD, Git, VSTS, JFrog Artifactory, Nexus
**Observability & Monitoring**: Prometheus, Grafana, Alertmanager, Splunk, Zabbix, CloudWatch, PagerDuty (On-call)
**Security & Quality**: Sonatype, Veracode, SonarQube, AWS WAF, Cloudflare WAF, External Secrets Operator
**Scripting & Development**: Python, Bash/Shell, PowerShell, TypeScript, Node.js, Next.js, React, SQL, FFmpeg

## Technical Projects & DevOps POCs

**GitOps Multi-Cluster EKS Platform Blueprint** | *Terraform, ArgoCD, Helm, AWS IRSA* DevOps Infrastructure POC
- Architected an automated multi-environment EKS deployment blueprint using modular Terraform and EC2 Launch Templates (AL2023).
- Configured GitOps application delivery with ArgoCD and Helm, integrating automated drift detection and health checks.
- Implemented least-privilege pod security via AWS IRSA and automated secret synchronization using External Secrets Operator.

**Thatiflix Converter** | *Electron, Node.js, FFmpeg, Shell* Desktop Application (macOS & Windows)
- Built cross-platform desktop application bundling native FFmpeg binaries for fast local video remuxing and stream-copying.
- Eliminated cloud upload latency for heavy media files through local multi-threaded process execution and progress tracking.

**Shift Rota Tracker** | *Next.js, React, Firebase, Capacitor* Cloud & Mobile Application
- Architected full-stack shift scheduling app with Firestore and Auth, deployed for web and mobile (iOS/Android).
- Integrated native background local reminders, offline state persistence, and real-time database synchronization.

**Passport Photo Maker** | *React, HTML5 Canvas API, Capacitor* Web & Mobile Image Processing
- Built client-side biometric photo utility with dynamic canvas guidelines adhering to official ICAO and UK passport specifications.
- Engineered local aspect ratio calculation, high-resolution cropping, and contrast scaling with zero server upload for privacy.

## Education

**Teesside University** Middlesbrough, UK
*Master of Science (MS) in Computer Science* *Jan. 2021 – Jan. 2023*
**Jawaharlal Nehru Technological University (JNTUH)** Hyderabad, India
*Bachelor of Technology (BTech)* *Aug. 2012 – May 2016*